

ELCT 361: Electromagnetics-2022-Spring

Quiz 4

Student Name:

Student ID or Matric Number:

< Solutions >

Q1. [50 pts] A coaxial conductor is shown in Fig. Q1. The length of conductor is l and the radii of inner conductor, and outer conductor are a , and b , respectively. The current I is flowing through the cylinder of inner conductor.

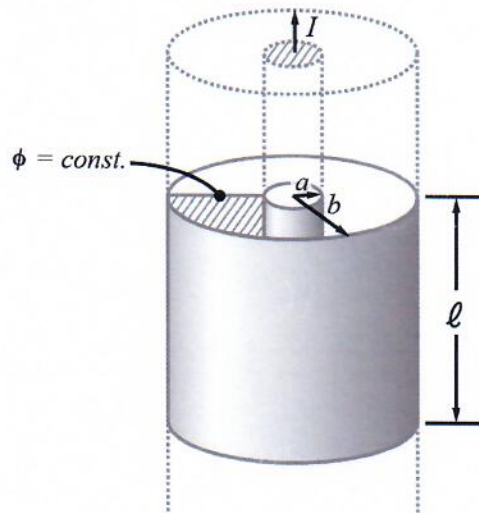


Fig. Q1 A coaxial conductor with inner and outer conductors.

- [10 pts] Find the $\vec{H}$ (magnetic field intensity) between the conductors, ($a < r < b$).
- [10 pts] If the space between the conductors is free space, find mmf and $\vec{B}$ ($a < r < b$)
- [10 pts] Find the magnetic flux, Φ . ($a < r < b$)
- [10 pts] Find the inductance per unit length of a coaxial conductor in Fig. Q1 and write the unit.
- [10 pts] Find the magnetic reluctance, $\mathcal{R}$.

Using Ampere's Circuital law, $\nabla \times \vec{B} = \mu_0 \vec{J}_c$ or $\nabla \times \vec{H} = \vec{J}_c$

(a) $a < r < b$, $\nabla \times \vec{H} = \vec{J}$ $\rightarrow \int_s (\nabla \times \vec{H}) \cdot d\vec{s} = \oint_c \vec{H} \cdot d\vec{l} = \int_s \vec{J} \cdot d\vec{s}$, considering Fig. Q1

$$\vec{H} = H_\phi \hat{a}_\phi \quad d\vec{l} = \hat{a}_\phi r d\phi$$

$$\oint_c H_\phi \hat{a}_\phi \cdot \hat{a}_\phi r d\phi = H_\phi \int_0^{2\pi} r d\phi = H_\phi 2\pi r = \int_s \vec{J} \cdot d\vec{s} = I$$

Therefore, $H\phi = \frac{I}{2\pi r} \rightarrow \underline{\underline{\vec{H} = \hat{a}_\phi \frac{I}{2\pi r}}}$

(b) In free space ($a < r < b$), $\mu_0 \quad \underline{\underline{\vec{B} = \mu_0 \vec{H} = \hat{a}_\phi \frac{\mu_0 I}{2\pi r}}}$, so, $\underline{\underline{mmf = \mu_0 I}}$

(c) $\underline{\underline{\Phi = \int_S \vec{B} \cdot d\vec{s} = \int_S \frac{\mu_0 I}{2\pi r} \hat{a}_\phi \cdot \hat{a}_\phi ds_\phi = \int_S \frac{\mu_0 I}{2\pi r} dr dz}}$
 $= \frac{\mu_0 I}{2\pi} \int_a^b \frac{1}{r} dr \int_0^l dz = \underline{\underline{\frac{\mu_0 I l}{2\pi} \ln\left[\frac{b}{a}\right] \text{ [T}\cdot\text{m]} or [Wb]}}$

(d) $L = \frac{\Lambda}{I}$, L per unit length, $\frac{L}{l} = \frac{\Lambda}{Il} = \frac{\Phi}{Il}$

Therefore $\frac{L}{l} = \frac{\Phi}{Il} = \frac{\frac{\mu_0 I l}{2\pi} \ln\left[\frac{b}{a}\right]}{Il} = \underline{\underline{\frac{\mu_0}{2\pi} \ln\left[\frac{b}{a}\right] \text{ [H/m]}}}$

(e) $\mathcal{R} = \frac{l}{\mu_0 S}$, considering Fig. Q1,

$S = \int_S ds = \int_a^b dr \int_0^l dz = (b-a)l$

Therefore, $\mathcal{R} = \frac{l}{\mu_0 (b-a)l} = \underline{\underline{\frac{1}{\mu_0 (b-a)}}}$

THE END